



STRICTLY CONFIDENTIAL

PREPARED FOR

Phase1 Factory Audit

100 kW

COMMERCIAL SOLAR PLANT

Phasel Factory Audit

Bhopal, MP On-Grid Commercial Solar 100 KW • 186 PANELS • 100% LOAD COVERAGE

INVESTMENT Gross ₹85.0 L Subsidy -₹78 k Net ₹84.2 L ₹84 k/kW • incl. GST

KEY PERFORMANCE INDICATORS

ANNUAL SAVING

₹10.2 L

25-YEAR ROI

12.1%

NET INVESTMENT

₹84.2 L

Gross ₹85.0 L • Subsidy -₹78 k

25-YEAR PROFIT

₹4.8 Cr

PAYBACK

8.3 yr

DCR vs Non-DCR — Cost Impact

Waaree 540W modules — subsidy & ALMM impact included

DCR / ALMM Higher subsidy potential ₹42,18,480 ₹42/Wp · 186 modules	NON-DCR Lower module rate ₹38,16,720 ₹38/Wp · 186 modules
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 DCR premium ₹4,01,760 (+11%) panel hardware vs Non-DCR	DCR / ALMM-listed modules may qualify for higher central subsidy under applicable schemes.
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Plant Capacity Scenarios

Alternative kW options alongside our recommended sizing

Recommended

☆ Recommended

100 kW

Net investment

Annual generation

Annual saving

Payback

Roof area (est.)

₹84,22,000

1,14,975 kWh

₹10,20,000

8.3 yr

~650 m²

Conservative

80 kW

Net investment

Annual generation

Annual saving

Payback

Roof area (est.)

₹67,37,600

91,980 kWh

₹8,16,000

8.3 yr

~520 m²

Expansion

120 kW

Net investment

Annual generation

Annual saving

Payback

Roof area (est.)

₹1,01,06,400

1,37,970 kWh

₹12,24,000

8.3 yr

~780 m²

Investment Analysis Dashboard

Full financial return profile — payback, IRR, and 25-year wealth creation

 FIRST-YEAR ROI

12.1%

initial yield on investment

 ESTIMATED IRR

8.7%

internal rate of return

 BREAK-EVEN YEAR

7

full investment recovery

 25-YEAR NET PROFIT

₹4.75 Cr

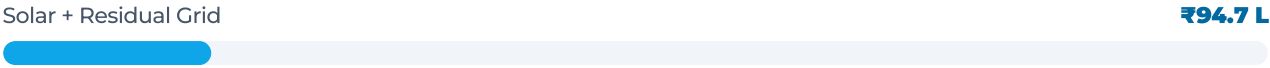
at 6% tariff escalation p.a.

CUMULATIVE CASHFLOW — 25 YEARS

Break-even at Year 7



25-YEAR ENERGY COST COMPARISON



NET LIFECYCLE SAVING

₹4.78 Cr

grid vs solar over 25 years

ENVIRONMENTAL ROI



25-Year Economic Analysis

Rising grid tariffs (6% p.a.) vs flat solar cost — detailed year-by-year cashflow

FIRST-YEAR ROI

12.1%

annualised

ESTIMATED IRR

8.7%

Rule-of-72 basis

LCOE

₹2.93/kWh

levelised cost of energy

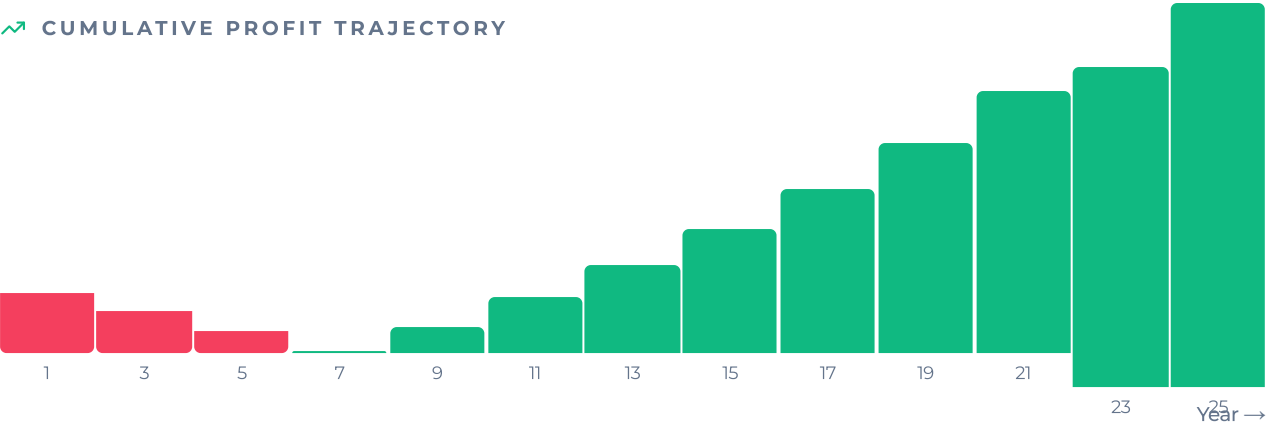
NPV (@ 10%)

₹69.8 L

positive

YEAR-BY-YEAR CASHFLOW STATEMENT			
YEAR	GRID COST	SOLAR SAVING	CUMULATIVE
Yr 1	₹12.0 L	₹10.2 L	₹-74.0 L
Yr 2	₹12.7 L	₹10.8 L	₹-63.2 L
Yr 3	₹13.5 L	₹11.5 L	₹-51.7 L
Yr 4	₹14.3 L	₹12.1 L	₹-39.6 L
Yr 5	₹15.1 L	₹12.9 L	₹-26.7 L
Yr 10	₹20.3 L	₹17.2 L	+₹50.2 L
Yr 15	₹27.1 L	₹23.1 L	+₹1.53 Cr
Yr 20	₹36.3 L	₹30.9 L	+₹2.91 Cr
Yr 25	₹48.6 L	₹41.3 L	+₹4.75 Cr
* Assumes 6% annual electricity tariff escalation			

 CUMULATIVE PROFIT TRAJECTORY



BREAK-EVEN SUMMARY

Full investment recovered in Year 7

Then **₹4.75 Cr** net profit over 25 years

Technical Design Basis

Key engineering parameters governing system sizing, layout, and performance estimates

SYSTEM DESIGN PARAMETERS

DC Capacity (STC)	100.44 kWp	AC Capacity (inverter rated)	100 kW
DC/AC Ratio	1	Peak Sun Hours (India avg)	5.0 hrs/day
Performance Ratio (PR)	78%	Capacity Factor (CF)	13.1%
Specific Yield	1150 kWh/kWp/yr	Annual Generation	115.0 MWh/yr
Panel Efficiency (STC)	~25.7%	Module Wattage	540 Wp per panel (Mono PERC)
Panel Count	186 nos	Load Coverage	100%

STANDARDS & COMPLIANCE

- IEC 61215 / BIS certified PV modules
- MNRE-ALMM listed panel brand
- IEC 62109 certified string inverter
- IS 13947 / IEC 60947 switchgear
- IS/IEC 62446 commissioning test
- IS 732 electrical installation wiring
- CEA Safety Regulations 2010
- Net-metering per SERC regulations

INSTALLATION & COMMISSIONING PROCESS

- 01 Site Survey

Shading analysis · Load profiling · Structural assessment
- 02 System Design

PVsyst simulation · SLD drawing · Structural design
- 03 Procurement

ALMM-listed panels · Certified inverter · BOS materials
- 04 Installation

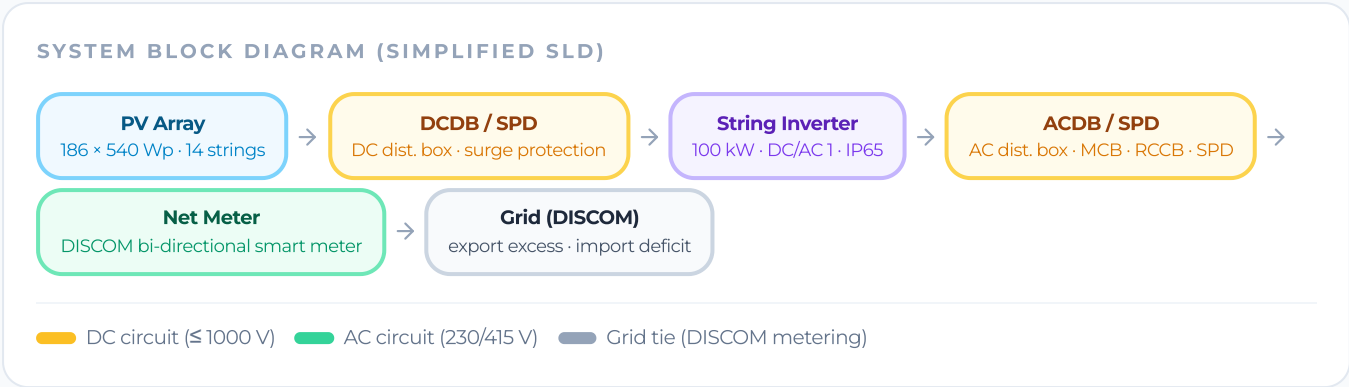
Certified engineers · Safe wiring · Grid interconnect
- 05 Testing & Commissioning

IV curve · Insulation test · Commissioning report
- 06 Net-Metering

DISCOM application · Meter installation · Grid export

System Component Architecture

DC string to grid — single-line diagram overview with component specifications



COMPONENT SPECIFICATIONS				
COMPONENT	SPECIFICATION	BRAND / STANDARD	QTY	WARRANTY
Solar PV Modules	540 Wp Mono PERC	JSW	186 nos	25 yr linear power
String Inverter	100 kW Grid-Tie, IP65	Growatt / Deye (BIS-certified)	1 nos	5 yr standard
Mounting Structure	Hot-dip galvanised MS, wind-load engineered	IS 2062 steel	100 kW capacity	10 yr structural
DCDB / ACDB	IP54 weatherproof, SPD + MCB + Isolator	Havells / Schneider / Legrand	1 set each	1 yr product
DC Cables	4 mm ² / 6 mm ² Solar DC (TÜV-certified)	Havells / Polycab	Per layout design	25 yr UV rating
Earthing System	IS 3043 — 4-electrode GI pipe earthing	Copper-bonded / GI	As per design	Lifetime

PROTECTION PROVISIONS

- DC circuit breakers + fuses (per string)
- Type-2 surge protection — DC & AC side
- RCCB + RCBO for AC installation
- Insulation monitoring relay (IMR)
- IS 3043 compliant 4-electrode earthing
- Anti-islanding protection per CEA

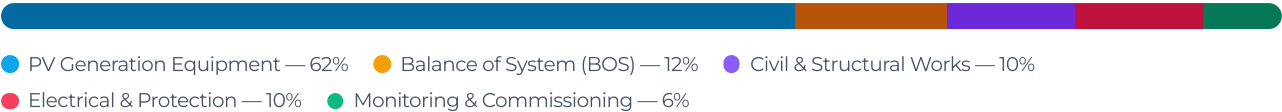
MONITORING & CONTROL

- Inverter cloud portal via Wi-Fi / LAN
- Real-time generation + consumption tracking
- String-level performance data
- Fault alarm + SMS / email alerts
- Mobile app — iOS & Android
- Monthly performance report (PDF)

Tiered Bill of Materials

Complete EPC supply-and-install scope — all categories with specifications, makes, and warranties

APPROXIMATE COST DISTRIBUTION



Total project value: ₹85.0 L · Net after subsidy: ₹84.2 L

DCR vs Non-DCR Panel Pricing 186 modules · 540W · Mono PERC

DCR PANEL

ALMM listed · Subsidy eligible

₹42.0/Wp

₹42.18 L total

PM Surya Ghar subsidy applicable

NON-DCR PANEL

Import / non-ALMM

₹38.0/Wp

₹38.17 L total

No subsidy · Lower upfront cost

➔ DCR Premium

+₹4.0/Wp

+₹402k total · offset by subsidy benefit

● PV GENERATION EQUIPMENT

ITEM	SPECIFICATION	MAKE / STANDARD	QTY	UNIT	WARRANTY
Solar PV Modules	540 Wp Mono PERC, $\eta \geq 20.8\%$	JSW	186	nos	25 yr linear
String Inverter (Three-phase)	100 kW on-grid, Three-phase, MPPT, IP65, LCD	Growatt / Deye (BIS-certified)	1	nos	5 yr standard

● BALANCE OF SYSTEM (BOS)

ITEM	SPECIFICATION	MAKE / STANDARD	QTY	UNIT	WARRANTY
Solar DC Cable (4 mm ²)	TÜV 2Pfg 1169, UV-rated, 1 kV	Polycab / Havells	As per required	metres	25 yr UV
AC Power Cable (16 mm ² / 25 mm ² / 35 mm ²)	XLPE armoured, IS 1554	Finolex / KEI	As per required	metres	10 yr
MC4 Connectors	IP68, TÜV certified, UV-stable	Stäubli / Amphenol	56	pairs	25 yr
Cable Management	25 mm / 40 mm IS 9537, UV-rated	Finolex / Sudhakar	As per layout	lot	10 yr

● CIVIL & STRUCTURAL WORKS

ITEM	SPECIFICATION	MAKE / STANDARD	QTY	UNIT	WARRANTY
Module Mounting Structure	Hot-dip galvanised MS / Aluminium mono rail, wind speed 120	IS 2062, zinc $\geq$ 85 μ m	100 kW	capacity	10 yr structural
Foundation / Anchor bolts	M12/M16 SS 304, chemical anchor	Fischer equiv.	372 approx	nos	10 yr
Cable Tray / Ladder	GI 2.0 mm perforated, IS 2062	Legrand equiv.	As per layout	lot	10 yr

● ELECTRICAL & PROTECTION

ITEM	SPECIFICATION	MAKE / STANDARD	QTY	UNIT	WARRANTY
DCDB (DC Distribution Box)	IP54, DC MCB + fuse + SPD Type 2	Havells / Schneider	1	nos	1 yr
ACDB (AC Distribution Box)	IP54, MCB + RCCB + SPD + energy meter	Havells / Schneider	1	nos	1 yr
Earthing System	IS 3043 — 4-electrode GI pipe, 3 m deep	Copper-bonded / GI	4 nos	electrodes	Lifetime
Lightning Arrestor	IEC 62305 — Class I/II, roof-mount	OBO Bettermann / equiv.	1 set	lot	5 yr

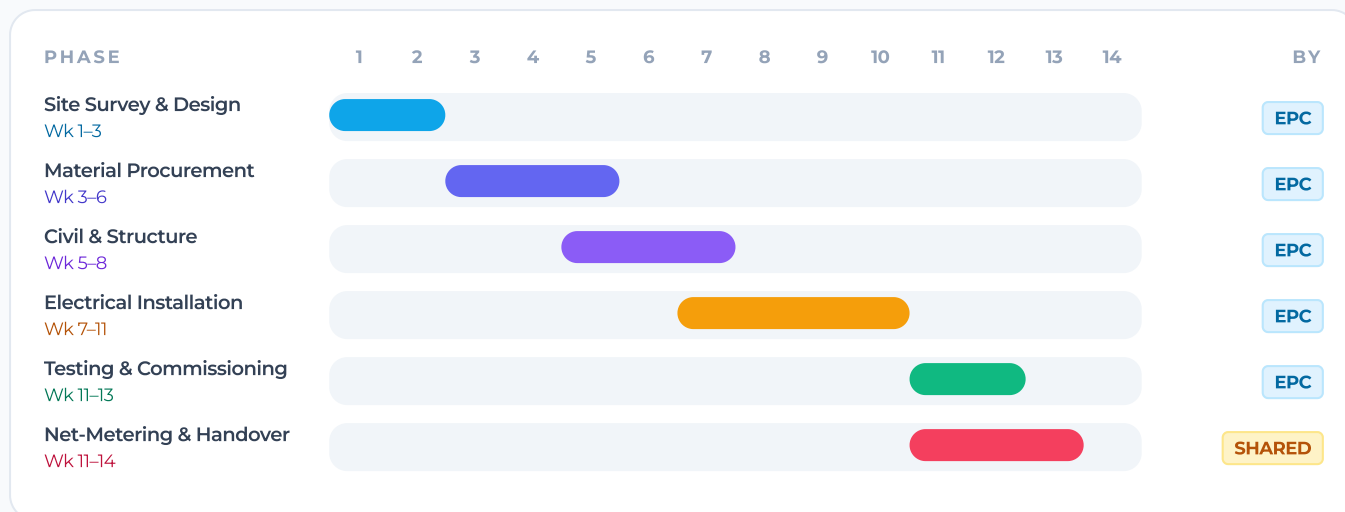
● MONITORING & COMMISSIONING

ITEM	SPECIFICATION	MAKE / STANDARD	QTY	UNIT	WARRANTY
Smart Monitoring Device	IoT data logger, Wi-Fi + Ethernet	Inverter OEM cloud portal	1	lot	2 yr
Net Energy Meter (NEM)	Bi-directional, CT/PT type per DISCOM spec	DISCOM approved make	1	nos	DISCOM owned
Commissioning & Testing	IV curve · insulation test · IS/IEC 62446 report	In-house certified engineers	1	lot	N/A
DISCOM Net-Metering Approval	Application + inspection + meter install support	EPC scope	1	application	N/A

☞ This BOM reflects indicative specifications. Final BOM will be provided post site survey and detailed design. Equivalent brand substitutions may be used with prior written approval.

Project Execution Timeline

Estimated 14-week project plan for 100 kW on-grid system



01 Site Survey & Design

- Shadow & tilt analysis
- Load profiling & tariff study
- PVsyst / SketchUp modelling
- SLD + structural drawing

🕒 Design approved

02 Material Procurement

- PO to panel & inverter vendors
- BOS items sourcing
- Logistics to site

🕒 Materials at site

03 Civil & Structure

- Structural foundation
- Module mounting structure erection
- Cable trays & conduit

🕒 Structure complete

04 Electrical Installation

- Panel mounting & wiring
- DCDB / ACDB installation
- Inverter & earthing
- AC interconnection

🕒 Pre-commissioning ready

05 Testing & Commissioning

- IV curve test
- Insulation resistance test
- System energisation
- IS/IEC 62446 commissioning report

🕒 System live

06 Net-Metering & Handover

- DISCOM NEM application
- Meter inspection & approval
- Training — customer team
- Document handover

🕒 Project closed

Smart Monitoring & Annual Maintenance

24×7 real-time data, proactive fault management, and long-term performance assurance

SMART MONITORING CAPABILITIES

-  **Real-Time Generation**
Minute-level kW output & kWh energy tracking
-  **String-Level Monitoring**
Per-string performance & mismatch detection
-  **Grid Export / Import**
Net-meter data + consumption analysis
-  **Fault Alarms & Alerts**
SMS / email — fault, low production, grid error
-  **Mobile App**
iOS & Android — live dashboard & history
-  **Monthly Performance Report**
PDF report — generation, saving, CO₂ data
-  **Historical Data**
5+ year historical data storage on cloud
-  **Remote Diagnostics**
Remote firmware update & inverter diagnostics

SERVICE LEVEL COMMITMENTS (SLA)

METRIC	STANDARD	PREMIUM
Response Time (remote)	4 business hrs	2 business hrs
On-site Visit (fault)	48 hrs	24 hrs
Inverter Downtime Cap	72 hrs max	48 hrs max
System Uptime Guarantee	≥ 95%	≥ 98%
Annual Service Visits	1 per year	2 per year
Report Delivery	Monthly	Weekly + Monthly

ANNUAL MAINTENANCE CONTRACT — PLAN COMPARISON

1 Year

Complimentary

First year — complimentary

- ✓ Annual inspection visit
- ✓ Panel cleaning (1 visit)
- ✓ Inverter firmware update
- ✓ Terminal tightening check
- ✓ Performance report
- ✗ Quarterly panel cleaning
- ✗ Spare parts coverage

RECOMMENDED

5 Year

₹1.7 L

15% discount — 5-year package

- ✓ Annual inspection visit
- ✓ Panel cleaning (2× / yr)
- ✓ Inverter firmware update
- ✓ Terminal tightening check
- ✓ Performance report
- ✓ Quarterly panel cleaning
- ✗ Spare parts coverage

10 Year

₹3.0 L

25% discount — 10-year package

- ✓ Annual inspection visit
- ✓ Panel cleaning (2× / yr)
- ✓ Inverter firmware update
- ✓ Terminal tightening check
- ✓ Performance report
- ✓ Quarterly panel cleaning
- ✓ Spare parts coverage

Commercial & Contractual Terms

Payment schedule, warranty matrix, EPC scope, and key commercial terms

 **PAYMENT SCHEDULE**

Total Project Value

₹84.2 L

1

advance

₹21.3 L (25%)

2

material

₹42.5 L (50%)

3

installation

₹17.0 L (20%)

4

commissioning

₹4.3 L (5%)

 KEY COMMERCIAL TERMS	
Quotation Validity	30 days from date of issue
Payment Terms	Milestone-based (see above)
Project Completion	14 weeks from WO
Price Basis	Inclusive of current GST + freight
GST Rate	12% (solar energy equipment)
Dispute Resolution	Arbitration Act 1996, India

 **EPC SCOPE SUMMARY**

INCLUDES

- All EPC supply and installation works
- DISCOM NEM application & liaison support
- IS/IEC 62446 commissioning test report
- Cloud monitoring portal activation
- Operations & maintenance training
- First-year AMC (complimentary)

EXCLUDES

- DISCOM net meter cost (DISCOM-owned)
- Internal load re-cabling / DB upgrade
- Roof waterproofing / civil repairs
- Electricity connection augmentation
- PM Surya Ghar subsidy processing

 WARRANTY MATRIX

COMPONENT	DURATION	WARRANTOR	COVERAGE
Solar PV Modules — Product	10 years	Manufacturer	Manufacturing defects
Solar PV Modules — Power Output	25 years	Manufacturer	Linear degradation $\leq 0.45\%/yr$
String Inverter	5 years	Manufacturer	Standard product warranty
Mounting Structure	10 years	EPC / Fabricator	Corrosion & structural integrity
Electrical Workmanship	2 years	EPC	Installation workmanship guarantee
Monitoring System	2 years	OEM / EPC	Hardware & software

* Contact Harihar Solar for warranty claims. All warranties subject to normal maintenance and use conditions.

Accept This Proposal

Decision summary, acceptance section, and next steps to commission your solar system

EXECUTIVE DECISION SUMMARY

100 kW On-Grid Solar System — Phase1 Factory Audit

NET INVESTMENT

₹84.2 L

post-subsidy

ANNUAL SAVING

₹10.2 L

from year one

BREAK-EVEN

7 yr

full recovery

25-YEAR ROI

12.1%

annualised

IRR (EST.)

8.7%

internal rate

25-YEAR PROFIT

₹4.75 Cr

net profit

COST OF DELAY

Every month without solar costs **₹85 k** in foregone savings.

A 3-month delay = **₹2.5 L** lost permanently.

How to Pay

Milestone schedule and bank / UPI details for transferring project payments.

₹ PAYMENT SCHEDULE		
Total payable		₹84.2 L
1	advance 25%	₹21.3 L
2	material 50%	₹42.5 L
3	installation 20%	₹17.0 L
4	commissioning 5%	₹4.3 L

🏦 BANK DETAILS	
ACCOUNT NAME	Harihar Solar
ACCOUNT NUMBER	—
IFSC	—
BRANCH	—
UPI ID	—



Scan to view proposal

PROPOSAL ACCEPTANCE

By accepting this proposal, you agree to the EPC scope and commercial terms described above. Work order to be issued to proceed.

Client Signature

Name & Designation

Date

Proposal Ref: 32F6823D

EPC Authorised Signatory

Harihar Solar

Place

CONTACT YOUR EPC PARTNER**Harihar Solar** +91-9993322267 · harihar@solar.com Contact for email**This 100 kW system will offset 2357 tonnes of CO₂ over 25 years.**

Equivalent to planting 107136 trees.